# fn then_index_or_default

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `then_index_or_default` in `mod.rs` at commit f5bb719 (outdated[1]).

This postprocessor indexes into a vector or returns the default value of the type if the index does not exist.

## 1 Hoare Triple

### Precondition

#### Compiler-Verified

- Generic `T` implements trait `Default`

#### User-Verified

None

### Pseudocode

```
def then_index_or_default(
    index: usize,
) -> Function[Vec[T], T]:
    return Function.new(lambda x: x[index] if index < len(x) else T.default())
```

### Postcondition

**Theorem 1.1.** For every setting of the input parameters (`index, T`) to `then_index_or_default` such that the given preconditions hold, `then_index_or_default` raises an error (at compile time or run time) or returns a valid postprocessor. A valid postprocessor has the following property:

1. (Data-independent errors). For every pair of members $x$ and $x'$ in `input_domain`, `function`$(x)$, `function`$(x')$ either both raise the same error, or neither raise an error.

*Proof.* The function is infallible, so the function satisfies the data-independent errors property. Therefore the postcondition is satisfied. □

[1] See new changes with `git diff f5bb719..09aa680 rust/src/transformations/scalar_to_vector/mod.rs`